

CENG 515

Numerical Methods for Computer Graphics

Instructors:

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Schedule:

- Monday 13:30

Course Description:

In this course, recent advances in computer graphics and fundamental topics in numerical methods for computer graphics are discussed.

Course Objectives:

1. To examine inspiring topics in computer graphics
2. To explore open subjects in popular research fields in computer graphics
3. To obtain the basic literacy in numerical methods

Reference Books:

Numerical Recipes – The Art of Scientific Computing – 3rd Edition by William H. Press, Saul A. Teukolsky, William T. Vetterling and Brian P. Flannery.

Numerical Algorithms – Methods for Computer Vision, Machine Learning, and Graphics by Justin Solomon.

Numerical Methods for Scientists and Engineers by Richard W. Hamming.

Reading:

Hart, J. Sphere tracing: a geometric method for the antialiased ray tracing of implicit surfaces. *The Visual Computer* **12**, 527–545 (1996).

<https://doi.org/10.1007/s003710050084>

Tentative Schedule:

Week		
1	Introduction	Reading – Sphere Tracing
2	Root finding methods	
3	Introduction to Shaders	HW – Newton Fractals
4	Implicit Surfaces, Signed Distance Fields	

5	Sphere Tracing	
6	Computer Graphics recap	HW – Sphere Tracing
7	Interval Arithmetic	
8	Introduction to Neural Implicits	Reading – Spelunking in the deep
9	Fast marching methods	
10	Intro to Monte Carlo Methods – pde estimation	Reading – Monte Carlo PDE Keenan Crane
11	Presentations	
13		
14		
15		

Suggested Papers for Presentation:

John C. Hart (1996). Sphere tracing: a geometric method for the antialiased ray tracing of implicit surfaces. *The Visual Computer*, 12, 527-545.

Galin, E., Guérin, E., Paris, A. and Peytavie, A. (2020), Segment Tracing Using Local Lipschitz Bounds. *Computer Graphics Forum*, 39: 545-554.
<https://doi.org/10.1111/cgf.13951>

Sawhney, R., & Crane, K. (2020). Monte Carlo geometry processing: a grid-free approach to PDE-based methods on volumetric domains. *ACM Trans. Graph.*, 39(4).

Vincent Sitzmann, Julien N. P. Martel, Alexander W. Bergman, David B. Lindell, & Gordon Wetzstein (2020). Implicit Neural Representations with Periodic Activation Functions. *CoRR*, abs/2006.09661.

Park, J., Florence, P., Straub, J., Newcombe, R., & Lovegrove, S. (2019). DeepSDF: Learning Continuous Signed Distance Functions for Shape Representation. In *The IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*.

Sharp, N., & Jacobson, A. (2022). Spelunking the deep: guaranteed queries on general neural implicit surfaces via range analysis. *ACM Trans. Graph.*, 41(4).